Name

Maitreya Gupta

Personal

DOB: 02.10.2000
Hobbies: Karate, KickBoxing, Game Dev, Coding
Languages: Hindi, English, Bengali

Technical Skills

Programming:
Python, C++, Java, C
Web:
React, Flask, Django, HTML/CSS
Robotics:
ROS, ESP32, MediaPipe, YOLO
Tools:
Git, VS Code, Godot, Three.js

Social Links

GitHub
HackerRank
TryHackMe
LinkedIn
Portfolio Website

Achievements

• State-level Bronze Winner in KickBoxing
• Gold Winner in Ju Srijan (Tech Fest of Jadavpur University)

Objective

M.Tech student in Intelligent Automation & Robotics with strong expertise in Computer Vision, Robotics, Full-Stack Development, and Interactive Simulations. Seeking AI/ML, Robotics, and Software Engineering roles where I can apply my research and development skills.

Professional Experience

Programmer Analyst Intern(10/23/2023 to 12/27/2023), Cognizant
Offer Letter **View** *Resignation Letter* **View**

- Worked on enterprise-level software development projects as part of the Programmer Analyst team.
- Gained hands-on experience with industry-standard development practices and agile methodologies.
- Contributed to full-stack development tasks, collaborating with cross-functional teams.
- Participated in code reviews, debugging, and testing to ensure high-quality software delivery.

Education

2024–Present	**M.Tech (Intelligent Automation and Robotics)**, Jadavpur University CGPA: 8.47/10, YGP: 77.75%
2019–2023	**B.Tech (Electronics and Communication)**, St. Thomas College CGPA: 8.33/10
2017–2019	**Patha Bhavan** (12th: 65%)
2015–2017	**Patha Bhavan** (10th: 83.28%)

Selected Certifications

- Fullstack Development - Coding Ninjas
- Python (Basic) - HackerRank
- SQL (Basic) - HackerRank
- BSNL Telecom Training

Technical Assessments

- **TCS iON National Qualifier Test (NQT) 2026**
  - Cognitive Ability: 82.85% | Advanced Quantitative & Reasoning: 93.97%
  - Programming (Python): 72.76%
  - Official Scorecard

Projects

Full-Stack & Mobile Development

- **Kanban Connection Board** — MERN Stack, React, Node.js, Express.js, MongoDB, JWT, Hugging Face API, Three.js
  Live Demo **GitHub** **Admin Portal** **Admin GitHub** *(Backend available on request)*
  - Developed a full-stack Kanban productivity platform featuring secure user authentication, task management, and interconnected task visualization via a Connection Board.

- Implemented AI-powered chatbot assistance and intelligent task suggestions leveraging the Hugging Face API.
  - Added drag-and-drop task management for intuitive movement of tasks across workflow stages.
  - Built secure JWT-based authentication with persistent task storage and automatic task creation timestamps.
  - Developed an administrative dashboard for monitoring user activity, reviewing tasks, tracking creation times, and flagging inappropriate content.
  - Enhanced user experience with responsive UI designs and Three.js visual elements.
  - Designed collaborative board functionalities to enable multi-user teamwork.
- **Android Job Posting Platform** — Java, Kotlin, MERN Stack **Showcase**
  - Built a full-stack job platform with secure REST APIs using Node.js, Express, and MongoDB.
  - Integrated CSV bulk upload functionality for efficient job posting management.
  - Developed a responsive Android UI with seamless client-server integration.

Game Development & Simulation

- **DexterGallery** — React, Three.js, WebGL **Live Demo** **GitHub**
  - Built an immersive 3D interactive simulation platform for exploring diverse scientific domains, including Physics, Chemistry, Astronomy, Robotics, and Quantum Mechanics.
  - Developed a Discovery Hub with real-time 3D lab simulations, enabling hands-on virtual experimentation across multiple disciplines.
  - Designed modular lab environments (Radiation Physics, Optical Deflection, Semiconductor, Electromagnetism, Kinematics, Mathematics, and Computer Science) for educational and research purposes.
  - Designed real time 3D IoT labs interactive with the enviroment using OpenCV Python under IOTHub.
  - Implemented a React-based frontend with seamless navigation and responsive UI for an intuitive user experience.
- **Industrial Multi-Agent Robot Navigation Simulator** — Unity 6, C# **Demo Video**
  - Developed a server-authoritative multiplayer simulator featuring A*, Dijkstra, RRT, and NavMesh navigation algorithms.
  - Integrated support for Python-trained PyTorch models to execute custom navigation behaviors.
  - Implemented full multiplayer synchronization utilizing Netcode for GameObjects.
- **Inception Battlegrounds** — Roblox Studio, Lua **Play**
  - Designed a multiplayer combat experience with custom gameplay systems, progression mechanics, and UI.
  - Executed full lifecycle development from initial design to final deployment.

Robotics & AI

- **CRMSF: Vision-Based Gesture-Controlled Robotic Arm** — Python, TensorFlow, MediaPipe, YOLO, ROS, Gazebo **Demo**
  - Developed a real-time vision-based robotic teleoperation system for intuitive control of a 6-DOF robotic arm and dexterous gripper using monocular hand tracking.
  - Designed the **CNN Regressive Motion Stabilization Filter (CRMSF)** to reduce hand-tracking jitter and improve the stability and smoothness of robotic motion.
  - Integrated MediaPipe hand tracking, YOLO-based object perception, and ThreeJS WebGL simulation into an end-to-end gesture-to-actuation pipeline for low-latency robotic control.
  - Research manuscript currently under preparation.
- **MobiUNet-3D: Hybrid Brain Tumour Segmentation and Volumetric Reconstruction** — TensorFlow, Keras, Python **Live Demo**
  - Developed a lightweight hybrid deep learning framework with a shared MobileNetV2 encoder for simultaneous brain tumour classification and MRI segmentation, reducing redundant computation through multi-task learning.

- Designed a unified segmentation–classification architecture with dedicated task-specific heads, enabling efficient feature reuse while maintaining a compact model footprint.
- Implemented a volumetric reconstruction pipeline that converts segmented MRI slices into patient-specific STL models using the Marching Cubes algorithm for three-dimensional visualisation.
- Integrated preprocessing, data augmentation, and K-fold cross-validation for robust model evaluation. Research manuscript under review.

- **Robotics Car Project** **Drive**
  - Implemented IR sensor-based obstacle detection with real-time data streaming to backend.
  - Created a Godot Engine interface with Python for data visualization and processing.

Bachelors Final Year Project

- **Compact Wideband MIMO Antenna for 5G and IoT** **Report**
  - Designed and simulated a compact four-element octagonal microstrip MIMO antenna using **CST Studio Suite**.
  - Achieved **2.64–20 GHz** bandwidth, **>20 dB isolation**, **6.53 dB peak gain**, and **ECC < 0.01**.
  - Fabricated and validated prototype in an anechoic chamber.